# CLINICAL DATA MANAGEMENT PROJECT REPORT

## Clinical Trial Data Validation and Quality Check

**Project Type:** Academic / Portfolio
**Domain:** Clinical Data Management
**Tools Used:** SAS 9.4, Microsoft Excel, SAS Studio
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# 1. Introduction

Clinical Data Management (CDM) is an important part of the clinical trial process. The main purpose of CDM is to make sure that clinical trial data is collected, reviewed, cleaned, and maintained in a consistent and reliable manner.

For this project, I worked on a simulated clinical trial dataset to understand how clinical data can be checked for completeness, consistency, and subject-level matching.

The project mainly focused on identifying data issues between different clinical datasets and preparing validation results that could be reviewed by a Clinical Data Manager or Clinical SAS Programmer.

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# 2. Project Objective

The main objectives of this project were:

- To understand the basic clinical data management workflow.
- To work with clinical trial datasets using SAS 9.4.
- To check whether subjects are consistently present across datasets.
- To identify unmatched subjects.
- To perform basic data quality checks.
- To identify missing and inconsistent data.
- To document validation findings.
- To understand how data discrepancies can be reported and followed up.

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# 3. Dataset Used

The project used simulated clinical trial datasets representing commonly used clinical domains.

The major datasets used were:

| Dataset | Description |
|---|---|
| Demog | Demographics |
| VITALS | Vital Signs |
| LABS | Laboratory Data |
| AE | Adverse Events |
| Visit | Visits |

The datasets contained subject-level information such as:

- Subject ID
- Visit information
- Date
- Sex
- Age
- Vital sign measurements
- Laboratory results
- Adverse event information

The data was used only for learning and project demonstration purposes.

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# 4. Project Workflow

The overall workflow followed in the project was:

**Raw Clinical Data**

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**Data Import**

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**Dataset Inspection**

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**Subject-Level Validation**

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**Missing Data Checks**

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**Cross-Dataset Checks**

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**Discrepancy Identification**

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**Validation Results**

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**Documentation**

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# 5. Software and Tools

## SAS 9.4

SAS was used for:

- Importing datasets
- Reviewing datasets
- Sorting data
- Identifying duplicate records
- Comparing subject IDs
- Performing basic data validation
- Generating validation outputs

## Microsoft Excel

Excel was used for:

- Reviewing validation results
- Maintaining discrepancy information
- Preparing summary tables
- Documenting findings

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# 6. Data Preparation

Before performing validation checks, the datasets were reviewed to understand their structure.

The following activities were performed:

1. Imported the datasets into SAS.
2. Checked the variable names and data types.
3. Reviewed the number of observations.
4. Checked subject ID variables.
5. Reviewed dates and visit variables.
6. Checked for missing values.
7. Sorted datasets based on the subject identifier.

Example SAS code used for reviewing a dataset:

```
proc contents data=work.vitals;
run;

proc print data=work.vitals(obs=20);
run;
```

The `PROC CONTENTS` procedure was used to understand the structure of the dataset, while `PROC PRINT` was used to review sample records.

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# 7. Subject Matching Check

One of the important checks performed during the project was checking whether subjects available in one clinical dataset were also present in the reference dataset.

For this project, the DEMOG dataset was treated as the main subject-level reference dataset.

The subject IDs in DEMOG were compared with the IDs in other datasets such as VITALS, LABS, and AE.

This helped identify subjects that were present in one dataset but could not be matched with the reference dataset.

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# 8. Dataset Validation Results

The subject matching validation produced the following findings:

| Dataset | Validation Check | Result |
|---|---|---|
| VITALS | Unmatched Subjects | 2 found |
| LABS | Unmatched Subjects | 1 found |
| AE | Unmatched Subjects | Finding identified during validation |

The unmatched records were treated as discrepancies requiring review rather than immediately being considered errors.

For example, an unmatched subject could occur because of:

- Incorrect Subject ID
- Data entry mistake
- Missing subject record
- Dataset update issue
- Different subject identifier format

Therefore, each finding would need to be reviewed against the source data before being corrected.

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# 9. Missing Data Check

Missing values were reviewed in important variables.

Examples of variables checked included:

- Subject ID
- Visit
- Date
- Test/measurement
- Result
- Adverse event information

Missing data was reviewed because incomplete records can affect the interpretation and quality of clinical trial data.

A missing value was not automatically treated as an error. Depending on the variable and study requirements, some fields may legitimately be blank.

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## 10. Duplicate Record Check

Duplicate records were also considered as part of the data quality review.

For example, duplicate records could occur when the same combination of:

- Subject ID
- Visit
- Date
- Test

appears more than once when only one record is expected.

A basic SAS approach for reviewing duplicates is:

```
proc sort data=work.vitals
          out=work.vitals_sorted
          nodupkey
          dupout=work.vitals_duplicates;
    by subject_id visit date;
run;
```

The duplicate output can then be reviewed separately to determine whether the records are true duplicates or valid repeated measurements.

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## 11. Data Consistency Checks

Basic consistency checks were performed to identify unusual or inconsistent records.

Examples include:

### Subject ID consistency

The same subject should generally have a consistent identifier across datasets.

### Date consistency

Dates were reviewed for missing or unusual values.

### Visit consistency

Visit information was reviewed to identify unexpected or inconsistent visit values.

### Measurement review

Clinical measurements were reviewed for missing or obviously inconsistent values.

These checks are useful for identifying records that require further investigation.

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## 12. Discrepancy Management

The identified findings were documented rather than directly modifying the original clinical data.

A simple discrepancy log structure was used:

| Finding | Dataset | Issue | Status |
|---|---|---|---|
| 1 | VITALS | Unmatched subject | Open |
| 2 | VITALS | Unmatched subject | Open |
| 3 | LABS | Unmatched subject | Open |
| 4 | AE | Validation finding | Review required |

The status of a discrepancy can later be updated as:

- Open
- Under Review
- Resolved
- Not a Finding
- Closed

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## 13. Example Validation Approach

A simple validation approach used in the project was:

1. Select the reference dataset.
2. Extract the unique Subject IDs.
3. Compare Subject IDs with the other datasets.
4. Identify IDs that do not match.
5. Review the unmatched records.
6. Document the findings.
7. Determine whether further action is required.

This exercise helped me understand that clinical data validation is not only about writing SAS code. It also involves understanding the data, identifying potential issues, documenting findings, and communicating discrepancies clearly.

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# 14. Challenges Faced

During the project, I faced some basic challenges while working with the datasets.

## 1. Understanding dataset relationships

Initially, it was difficult to understand how different clinical datasets were related to each other.

## 2. Subject-level matching

Understanding how the same subject is tracked across multiple datasets required careful review of Subject IDs.

## 3. Data quality issues

Unmatched subjects and missing values required additional investigation instead of simply changing the data.

## 4. SAS programming

While working with SAS, I had to understand procedures such as:

- PROC CONTENTS
- PROC PRINT
- PROC SORT
- DATA step
- BY-group processing

These exercises helped improve my understanding of basic SAS programming.

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# 15. Skills Demonstrated

Through this project, I gained practical exposure to:

- Clinical Data Management concepts
- Clinical trial datasets
- Subject-level data validation
- Data cleaning concepts
- Data quality checks
- Missing data review
- Duplicate checking
- SAS 9.4

- Basic SAS programming
- Excel-based discrepancy tracking
- Clinical data documentation

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## 16. Learning Outcomes

This project helped me understand the basic responsibilities involved in clinical data review.

I learned that clinical data needs to be checked systematically before it can be considered reliable.

I also learned the importance of:

- Maintaining data consistency
- Identifying discrepancies
- Avoiding unnecessary changes to source data
- Documenting validation results
- Reviewing findings before considering them as errors

The project also gave me a better understanding of how SAS can be used as a supporting tool in clinical data management.

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## 17. Conclusion

The project provided practical exposure to basic clinical data validation activities using simulated clinical trial data.

The main focus was on subject matching, data quality checks, missing data review, duplicate checking, and discrepancy documentation.

The validation exercise identified unmatched subjects in VITALS and LABS and additional findings requiring review in AE.

Overall, the project helped me connect the theoretical concepts of Clinical Data Management with practical data review using SAS 9.4.

As a fresher, this project also helped me develop a basic understanding of how clinical data is reviewed before being used for further clinical reporting and analysis.

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# 18. References


1. SAS 9.4 Documentation – SAS Programming concepts.
2. CDISC – Clinical Data Standards.
3. Basic Clinical Data Management concepts and study materials.
4. Good Clinical Practice (GCP) principles.